

HERITAGE TREASURES : AN IN-DEPTH ANALYSIS OF UNESCO WORLD HERITAGE SITES

Project Overview

Introduction

UNESCO World Heritage Sites represent globally significant cultural, natural, and mixed heritage locations. The dataset contains site names, countries or states, regions, categories, danger status, inscription years, area, latitude, and longitude. Manual analysis makes it difficult to identify concentration patterns, endangered sites, and long-term inscription trends. This project uses Tableau for visual analytics and Python Flask for web presentation.

Problem Statement

Global heritage data contains many records and analytical dimensions. Without effective visualization, comparing countries, identifying sites at risk, understanding regional inscription trends, and examining geographic distribution becomes difficult. An analytical solution is required to present UNESCO heritage information in a simple, visual, and user-friendly format.

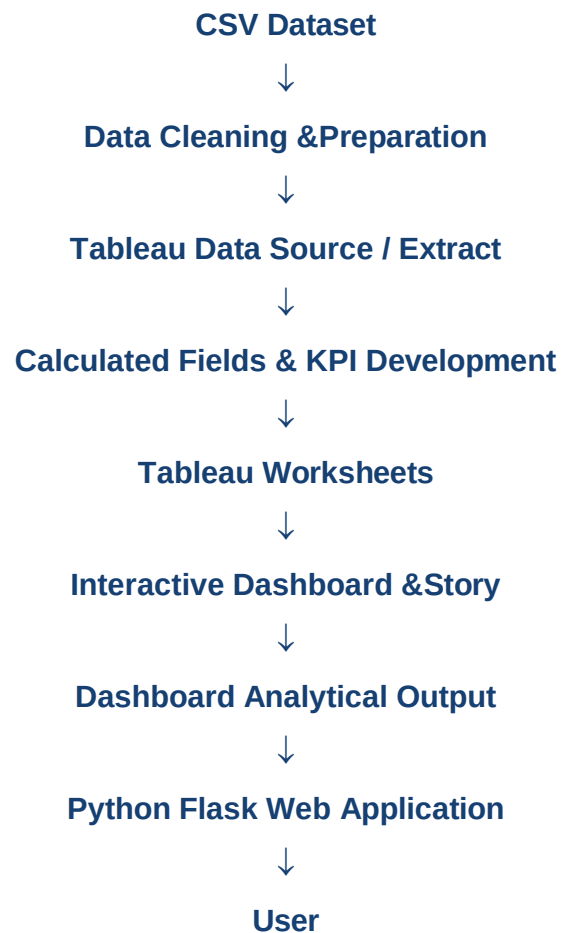
Objectives

- Analyze UNESCO World Heritage Sites by country or state.
- Identify and quantify heritage sites classified as being in danger.
- Study regional inscription trends from 1978 to 2019.
- Visualize global geographic distribution using latitude and longitude.
- Create KPI cards, analytical charts, a Tableau dashboard.
- Present the analytical output through a Python Flask web application.

Features of the Project

Data Cleaning & Preparation • KPI Analysis • Country/State Treemap • Heritage Risk Analysis • Regional Trend Analysis • Global Heritage Map • Dashboard Filters • Tableau Dashboard • Tableau Story • Performance Validation • Flask Web Integration • Responsive User Interface

System Architecture



The architecture follows a visualization-driven analytics workflow. The UNESCO CSV dataset is prepared and connected to Tableau. Analytical worksheets are created for KPIs, country distribution, heritage risk, inscription trends, and global mapping. These visualizations are combined into a dashboard and story. The final dashboard output is presented inside a Flask-based frontend for browser access.

Technologies Used

Technology	Purpose
CSV Dataset	Source data for UNESCO World Heritage Site analysis
Tableau	Calculated fields, KPIs, charts, maps, dashboard, and story
Python Flask	Web application and frontend routing
HTML	Structure Home, Dashboard, and About pages
CSS	Responsive dark-blue Heritage Insights interface
GitHub	Version control and repository hosting

Working of the Project

Step1–DataCollection

The UNESCO dataset is collected in CSV format with site, country/state, region, category, danger, year, area, latitude, and longitude fields.

Step2–DataPreparation

The dataset is reviewed for missing values, field types, duplicates, and analytical readiness.

Step3–TableauDataSetup

The prepared data is connected to Tableau and required dimensions, measures, geographic fields, and calculated fields are configured.

Step4–VisualizationDevelopment

KPI cards, country/state treemap, heritage risk donut chart, regional trend line chart, and global heritage map are developed.

Step5–Dashboard&Story

Worksheets are combined into the UNESCO Global Heritage Insights dashboard. Filters and actions support analysis, while story points present findings in sequence.

Step6–FlaskIntegration

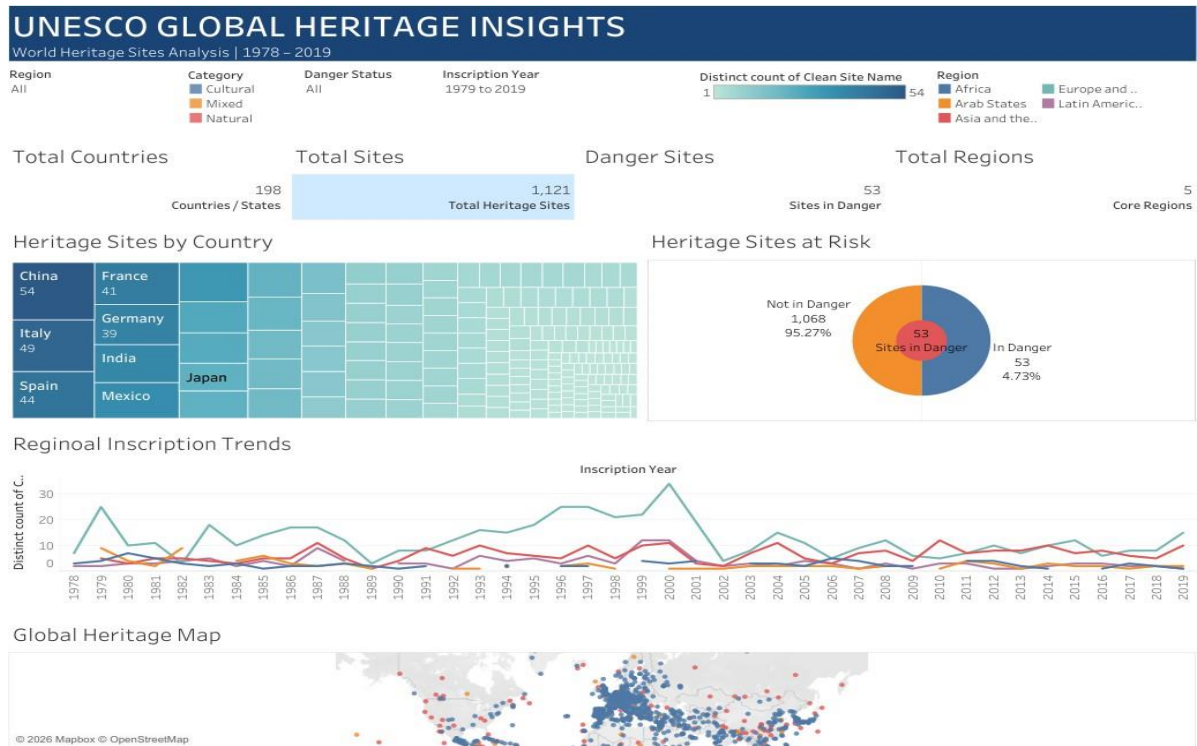
A Flask application is developed with Home, Dashboard, and About pages. The dashboard output is displayed through the frontend.

Step7–GitHubRepository

Source code, templates, stylesheet, dashboard image, requirements, and documentation are prepared for GitHub submission.

Result

The project successfully analyzes 1,121 UNESCO World Heritage Sites across 198 countries/states and five core regions. The dashboard identifies 53 sites in danger, representing 4.73% of the analyzed heritage portfolio. The treemap highlights heritage-rich countries, the risk chart compares endangered and non-endangered sites, the line chart shows regional inscription trends from 1978 to 2019, and the world map presents global geographic distribution.



The Flask frontend provides a professional Heritage Insights interface with navigation to the dashboard and project information. The solution demonstrates an end-to-end analytics workflow from raw dataset preparation to Tableau visualization and web presentation.

Conclusion

This project demonstrates how Tableau can transform raw UNESCO heritage data into meaningful visual insights. The dashboard makes it easier to understand global heritage concentration, identify preservation priorities, compare regional patterns, and examine inscription trends. Flask integration adds a clean web presentation layer, making the analytical output easy to access and suitable for internship demonstration and project submission.

